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Original Articles

Novel indicator for assessing wetland degradation based on the index of hydrological connectivity and its correlation with the root-soil interface

Yinghu Zhang [a,c,d], Jinhong Chen [a,c,d], Jinchi Zhang [a,c,d], Zhenming Zhang [b], Mingxiang Zhang [b,*]

[a] Co-Innovation Center for Sustainable Forestry in Southern China, Nanjing Forestry University, Nanjing 210037, PR China
[b] School of Ecology and Nature Conservation, Beijing Forestry University, Beijing 100083, PR China
[c] Jiangsu Provincial Key Laboratory of Soil Erosion and Ecological Restoration, Nanjing Forestry University, Nanjing 210037, PR China
[d] Priority Academic Program Development of Jiangsu High Education Institutions (PAPD), Nanjing Forestry University, Nanjing 210037, PR China

A R T I C L E I N F O

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A B S T R A C T

In this study, an innovative method was used to assess the spatial and temporal patterns of the hydrological connectivity in soil profiles in the Yellow River Delta wetland. In this method, field dye-tracing experiments conducted in the study area (i.e., large [LWa] and small communities [LWb] of *Phragmites australis* and *Suaeda glauca* [JP]) were considered, and the root-soil interface in the region and the status of the wetland degradation were analyzed. The results revealed that the index of hydrological connectivity (*IHC*) significantly changed both spatially and temporally. The high *IHC* values reaching a 0.392 ± 0.209 gradient were concentrated in the middle and western parts of the study areas. The spatial distribution of the high hydrological connectivity has been extremely fragmented since 2010. The LWb (severely degraded wetland) and JP (extremely degraded wetland) were found to be more seriously degraded than the LWa (moderately degraded wetland) in the study area. The changes in the *IHC* were positively correlated with the principal component (PC) values of the wetland degradation. The *IHC* is a novel indicator of the status of wetland degradation. Furthermore, the soil holding capacity, soil non-capacity porosity, and soil ventilation were relatively important for the changes in the *IHC*. Compared with the soil properties, the hydrological responses of the roots systems can be neglected at the root-soil interface. Based on our results, we propose an alternative wetland restoration solution for the Yellow River Delta wetland: 1) a seepage layer in the surface soils (0–10 cm), shallow ploughing treatment, and litter or straw return to the soil surface should be conducted to increase the hydrological connectivity of the soil surface; 2) reeds should not be reaped every year to remove the nutrients from this area; and 3) appropriate freshwater inputs should be strengthened.

1. Introduction

Hydrological connectivity has emerged as a key concept across the disciplines of ecology, hydrology, and geomorphology (Zhang et al., 2021). The concept of hydrological connectivity is of importance for the assessment of healthy soil conditions, the restoration of degrade wetland, and the identification of early warning signals of wetland degradation (Rodríguez et al., 2018; Saco et al., 2020). In wetland ecosystems, hydrological connectivity is an important indicator used to determine the wetland's biodiversity, population richness, and other ecosystem services (Li et al., 2021a,b; Yang et al., 2021). Disruption of the hydrological connectivity, referred to as hydrological alterations, can degrade wetland ecosystems (Castello and Macedo, 2016). However, previous studies of the hydrological connectivity have primarily focused on using the hydrological connectivity between or within the wetland surface water bodies to reflect the wetland health (Castello and Macedo, 2016; Yang et al., 2021), whereas studies focused on using the hydrological connectivity in soil profiles to detect the status of wetland degradation is lacking.

Hydrological connectivity in soil profiles has been identified as being an important hydrological response below the soil surface, which regulates the efficiency of the water flow and solute transport (Zhang et al., 2021). Evidence suggests that the hydrological responses below the soil surface are more highly related to roots systems and soil properties

* Corresponding author.
E-mail address: pku2015hold@163.com (M. Zhang).

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(Zhang et al., 2017). The connected paths in soil profiles caused by the development of roots systems and soil properties are increasingly being recognized as important drivers of high hydrological responses. Recent studies have identified the root-soil interface as an important factor that controls soil and water conservation in dryland ecosystems and that ultimately influences the hydrodynamics (Rodríguez et al., 2018). The hydrodynamics determine the water redistribution conditions and the energy absorbed by roots systems, which have been shown to control the survival and production of wetland vegetation (Saco et al., 2020). Assessing the relationship between soil properties and hydrological connectivity is essential to establishing a scientific basis for the restoration of degraded wetlands (Feng et al., 2021). The spatial pattern of the root-soil interface is linked to the wetland ecosystem functions (Rodríguez et al., 2018). Therefore, maintaining wetland health may require understanding the changes in the hydrological connectivity in soil profiles caused by root systems and soil properties. However, predicting the impact of the root-soil interface on the hydrological connectivity in soil profiles is difficult due to the lack of available knowledge regarding hydrological connectivity in soil profiles.

Understanding the hydrological connectivity in soil profiles can help us identify the locations where wetland degradation is likely to be pronounced. However, little is known about the hydrological connectivity in soil profiles, which is important in terms of identifying its value range that is most susceptible to wetland degradation. Many wetland degradation studies have relied on different indicators to investigate whether the wetland in a given area is degraded or not (Turnbull and Wainwright, 2019). Accordingly, several indicators have been proposed as early warning signals of wetland degradation and for identifying ecosystem functions (Rodríguez et al., 2018). Consequently, different criteria have been used and very different rankings have been formulated (Servadio and Convertino, 2018). Although a number of indicators of wetland degradation have been developed, there is no international standard or integrated indicator that identifies the system's state (Shen and Liu, 2021). In this study, we constructed a novel indicator that reflects the status of wetland degradation. Using this indicator, we developed expressions for its value and variance under different degraded wetland conditions. Approaches for constructing such an indicator are uncommon in published papers, and the proposed indicator, which can provide an assessment of wetland degradation, is a new topic of interest.

The Yellow River Delta is one of the priority areas and is the one of the most active deltas based on its ecological and economic importance (Liu et al., 2019). During the last several decades, the delta has experienced changes, which have been attributed to agricultural exploration, expansion of the urban residential area, and petroleum refining. These environmental changes have caused the deterioration of the structure and function of the wetland ecosystems (Lu et al., 2021), especially wetland degradation (Meng et al., 2019; Deng, 2020). Wetland degradation in this area enhances the complexity and spatial heterogeneity between the isolated wetlands and the surrounding water bodies (Zhu et al., 2020). The shift in the landscape between *Suaeda glauca* dominated patches and vegetation patches dominated by *Phragmites australis* in the area has been promoted by the changes in the hydrodynamics. Given this trend, understanding the spatial and temporal distribution of the hydrological connectivity and determining the different factors of the hydrological connectivity are critically important to wetland management. In this study, we used field dye-tracing experiments 1) to characterize the hydrological connectivity both spatially and temporally; 2) to quantify the relationship between the hydrological connectivity in the soil profiles and the wetland degradation; and 3) to determine the integrated effects of the root-soil interface on the hydrological connectivity. Understanding the hydrological connectivity in soil profiles is the first step in wetland restoration. The novel indicator of the hydrological connectivity developed in this study is expected to be the foundation of future work. Knowledge of this novel indicator of wetland degradation in the study area can help other regions avoid similar problems and effectively improve wetland management.

2. Materials and methods

2.1. Study area description

The study area is located in the Yellow River Delta wetland (117°31′–119°18′E, 36°55′–38°16′310″N) in northeastern Shandong Province, China. The Yellow River Delta wetland covers an area of 5450 km^2. The main soil type in the Yellow River Delta wetland is coastal saline soil, which is derived from the sediments (Zhang et al., 2016). The three main vegetation types are deciduous broad-leaved forests, marsh vegetation, and salt-bearing vegetation. The dominant vegetation is *Robinia pseudoacacia* in the deciduous broad-leaved forest communities, *Phragmites australis* in the marsh vegetation communities, and *Tamarix chinensis* and *Suaeda salsa* in the salt-bearing vegetation communities. The climate is warm-temperate continental with a mean annual precipitation of 576.7 mm (70% of which occurs in July and August), a mean annual potential evapotranspiration of 1,962 mm, a mean annual temperature of 12.1 °C, and a frost-free period of 196 d.

The Yellow River Delta wetland is one of the most active regions worldwide. It is the youngest wetland ecosystem in China and plays a significant role in water purification, biodiversity conservation, and bird migration (Chen et al., 2016). In addition to its ecological functions, it also has economic development potential (Bai et al., 2019). However, it has been suffering from different degrees of wetland degradation, leading to the recession of the wetland ecosystem's functions due to human activities and climate change in recent years (Feng et al., 2021). The Yellow River Delta wetland has been on the list of Ramsar Wetlands of International Importance since 2013 (Zhao et al., 2019). The main wetland restoration in this area is designed to bring freshwater to the wetland and to resist saltwater intrusion (Cui et al., 2009). In the eastern part of the Yellow River Delta wetland, a National Nature Reserve was conducted for numerous experiments, which were designed to determine the mechanisms of wetland degradation. The reserve includes three areas, i.e., Huanghekou Station, Yiqian'er Station, and Dayangliu Station. The experimental plots in this study were established in the main landscape vegetation patches of Huanghekou Station (Fig. 1).

2.2. Sample collection

In the wet season from July to August 2017, experimental plots were typically chosen based on the main vegetation species. The samples were chosen in large [LWa] and small communities [LWb] of *Phragmites australis* and *Suaeda glauca* [JP] with flat topography. The vegetation species were investigated in the 1.0 × 1.0 m area of land surface at the same time. The LWa wetlands are areas with a higher above-ground dry weight (1000 g), coverage (98%), and height (2.0 m), whereas the LWb wetlands are typical areas with a lower above-ground dry weight (200 g), coverage (40%), and height (0.6 m). The JP wetlands are areas with an above-ground dry weight of 150 g, a coverage of 25%, and a height of 0.2 m. The soil properties and roots parameters of the LWa, LWb, and JP areas are presented in Table 1.

Soil samples were collected from each soil depth interval (0–10, 10–20, and 20–30 cm) in each experimental plot. The soil samples were transported to the laboratory immediately. One part of the soil samples was air dried at room temperature for one month. The samples were sieved through a 2 mm mesh to remove any stones. The soil samples were divided into two subsamples. One subsample was crushed and sieved through a 1 mm sieve to measure the soil pH and soil electronic conductivity (SEC) (Zhao et al., 2019). The soil pH was measured using a soil:water (1:5, m/v) slurry (Orion 3 Star, Thermo Electron) (Zhao et al., 2019). The SEC was measured using the conductometric method. Another subsample was sieved through a 0.149 mm mesh to determine the soil organic matter (SOM) and soil total nitrogen (STN) contents. The SOM content was analyzed using dichromate oxidation-colorimetric

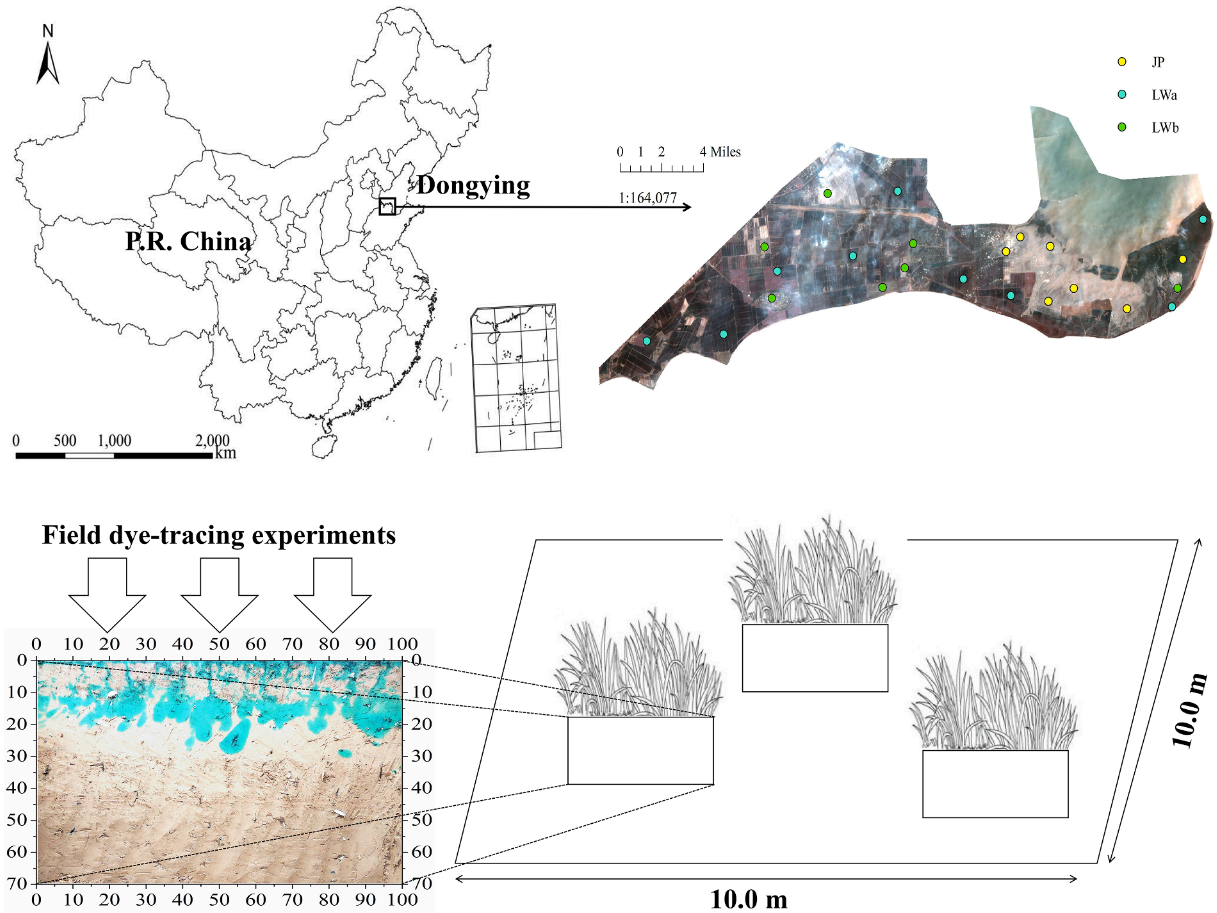

Fig. 1. Location of the study areas.

determination (Sims and Haby, 1971). The STN content was determined using the Kjeldahl method (Bremner, 1960). The soil samples were digested in Teflon tubes using a mixture of $HClO_4$, HNO_3, and HF for analysis of the soil total phosphorus (STP_1) content. The digested sample solutions were analyzed using inductively coupled plasma atomic emission spectrometry (Zhang et al., 2016). The soil available phosphorus (SAP) contents of all of the extracts were determined using the colorimetric method and 0.5 M $NaHCO_3$ solution (Olsen et al., 1954). Another part of the soil samples was collected to determine the soil total porosity (STP_2), soil non-capillary porosity (SNCP), soil maximum holding capacity (SMHC), soil capillary holding capacity (SCHC), soil optimum water content (SOWC), soil water storage (SWS), soil ventilation (SV), and soil drainage capacity (SDC) (Jiang et al., 2019). The soil samples were dried at 105 °C for 24 h in an oven to determine the soil water content (SWC) and the soil bulk density (SBD) (Zhao et al., 2019). Then, these soil samples were placed in dishes with 4–5 mm of water for 2 h so that the roots could spread and the soil particles could be easily removed, and then, the soil particles were separated from the roots under running water and 5 mm mesh sieves. The roots were dried for 48 h in an oven at 70 °C, and then, they were analyzed using the RootAnalysis professional software (PM-Tech GmbH, Germany) to determine the different roots parameters, i.e., the root length (RL), width (RW), surface area (RSA), projected area (RPA), volume (RV), and biomass (RB).

2.3. Field dye-tracing experiments

Field dye-tracing experiments (Fig. 1) with three replicates were conducted in the 10.0 × 10.0 m quadrat of each experimental plot to characterize the dye coverage (*DC*) and fractal dimension (*FD*) in the soil profiles (Fig. 2). The field methodology was resource-effective and had few restrictions regarding operation, as well as a low adsorption and high mobility (Guo et al., 2019). A Brilliant Blue FCF (C.I. food blue 2) dye solution (4 g/L) with a total volume of 50 L was uniformly applied to the 1.2 × 1.2 m area of land surface. Horizontal soil profile intervals of 10 cm based on the maximum dye depth were excavated one day after the application of the dye solution. The horizontal soil profiles were prepared for photographing using a digital camera before the sampling. The stained horizontal profiles were prepared for the processes of enhancing the image contrast, image de-noising, and image binarization using the ImageJ 1.8.0Fiji software (Yi et al., 2020). The pixels in each binarized image were counted using MATLAB. The *DC* and *FD* were determined using ImageJ.

- *DC* is defined as the percentage ratio of the dye-stained area to the total horizontal soil profile area.

$$DC = [D/(D + ND)]\cdot 100\% \tag{1}$$

where *D* is the dye-stained area, and *ND* is the non-dye-stained area.

- *FD* characterizes the complexity of the dye-stained area in the horizontal soil profiles. The larger the *FD* is, the more complex the dye-stained area is.

$$FD = \lim_{\varepsilon \to 0}[logN(\varepsilon)/log(1/\varepsilon)] \tag{2}$$

where ε is the average width of each dye-stained area, and $N(\varepsilon)$ is the number of fractals covered by this dye-stained area.

Table 1
Roots parameters and soil properties in the study areas.

Plots	SD (cm)	SBD (g/cm^3)	STP$_1$	SNP	SWC	SMHC	SV	SDC	SOM (g/kg)	STN (g/kg)	STP$_2$ (g/kg)	SAP (mg/kg)	pH	SEC (us/cm)	RL (mm/100 cm^3)	RW (mm/100 cm^3)	RSA (mm^2/100 cm^3)	RPA (mm^2/100 cm^3)	RV (mm^3/100 cm^3)	RB (g/100 cm^3)	VC (%)
LWa	0–10	0.933 ± 0.123	0.571 ± 0.026	0.024 ± 0.018	0.260 ± 0.058	0.626 ± 0.104	0.552 ± 0.002	0.189 ± 0.066	21.039 ± 9.265	1.345 ± 0.631	0.614 ± 0.033	3.927 ± 1.933	8.47 ± 0.18	114.2 ± 26.8	2443.99 ± 183.20	112.89 ± 19.07	4017.96 ± 305.71	1268.61 ± 100.34	2695.25 ± 579.60	0.45 ± 0.20	98
	10–20	1.308 ± 0.046	0.479 ± 0.013	0.010 ± 0.000	0.228 ± 0.027	0.367 ± 0.023	0.457 ± 0.040	0.031 ± 0.005	5.050 ± 0.503	0.358 ± 0.043	0.569 ± 0.055	2.398 ± 1.031	8.58 ± 0.24	145.5 ± 111.3	1620.34 ± 452.44	78.11 ± 17.68	2877.35 ± 636.60	906.68 ± 204.84	1945.89 ± 630.66	0.25 ± 0.07	
	20–30	1.404 ± 0.031	0.475 ± 0.020	0.007 ± 0.000	0.234 ± 0.022	0.339 ± 0.020	0.451 ± 0.012	0.020 ± 0.003	3.857 ± 1.257	0.270 ± 0.094	0.538 ± 0.045	1.792 ± 0.111	8.80 ± 0.19	209.1 ± 71.9	818.37 ± 781.80	31.51 ± 28.56	1019.54 ± 900.06	323.31 ± 285.27	307.43 ± 287.13	0.27 ± 0.09	
LWb	0–10	1.351 ± 0.015	0.487 ± 0.004	0.007 ± 0.001	0.236 ± 0.004	0.362 ± 0.001	0.446 ± 0.003	0.027 ± 0.005	11.244 ± 2.839	0.694 ± 0.038	0.694 ± 0.073	3.576 ± 0.279	8.96 ± 0.05	1160.0 ± 104.1	1743.73 ± 99.54	87.05 ± 25.81	3763.27 ± 1100.05	1188.02 ± 343.68	3059.05 ± 2146.97	0.47 ± 0.12	40
	10–20	1.346 ± 0.081	0.475 ± 0.016	0.004 ± 0.002	0.223 ± 0.014	0.355 ± 0.036	0.436 ± 0.016	0.029 ± 0.011	2.781 ± 0.693	0.200 ± 0.033	0.582 ± 0.003	1.749 ± 0.181	9.06 ± 0.25	770.7 ± 125.7	1044.90 ± 522.18	44.41 ± 12.92	2065.31 ± 1132.75	651.98 ± 358.47	1688.06 ± 1048.75	0.25 ± 0.03	
	20–30	1.424 ± 0.031	0.471 ± 0.018	0.005 ± 0.004	0.199 ± 0.052	0.331 ± 0.020	0.435 ± 0.007	0.020 ± 0.005	2.535 ± 1.072	0.159 ± 0.073	0.584 ± 0.021	1.552 ± 0.746	9.14 ± 0.17	712.3 ± 227.8	760.18 ± 314.52	37.58 ± 15.05	1889.46 ± 1365.76	600.10 ± 434.31	1304.06 ± 1067.70	0.18 ± 0.09	
JP	0–10	1.365 ± 0.091	0.489 ± 0.026	0.011 ± 0.008	0.243 ± 0.009	0.361 ± 0.042	0.445 ± 0.027	0.019 ± 0.008	6.219 ± 1.224	0.368 ± 0.060	0.551 ± 0.009	3.520 ± 0.069	8.37 ± 0.49	2191.3 ± 354.2	0	0	0	0	0	0	
	10–20	1.441 ± 0.002	0.455 ± 0.005	0.001 ± 0.000	0.235 ± 0.003	0.316 ± 0.003	0.411 ± 0.004	0.009 ± 0.001	2.400 ± 0.376	0.198 ± 0.020	0.563 ± 0.013	1.927 ± 0.869	9.02 ± 0.14	1838.7 ± 16.8	0	0	0	0	0	0	25
	20–30	1.417 ± 0.056	0.460 ± 0.007	0.002 ± 0.001	0.238 ± 0.009	0.324 ± 0.016	0.416 ± 0.006	0.010 ± 0.002	3.294 ± 0.545	0.217 ± 0.020	0.622 ± 0.079	2.365 ± 0.036	9.01 ± 0.05	2003.7 ± 758.9	0	0	0	0	0	0	

SD: Soil depth, SBD: Soil bulk density, STP$_1$: Soil total porosity, SNCP: Soil non-capillary porosity, SWC: Soil water content, SMHC: Soil maximum holding capacity, SV: Soil ventilation, SDC: Soil drainage capacity, SOM: Soil organic matter, STN: Soil total nitrogen, STP$_2$: Soil total phosphorus, SAP: Soil available phosphorus, SEC: Soil electronic conductivity, RL: Root length, RW: Root width, RSA: Root surface area, RPA: Root projected area, RV: Root volume, RB: Root biomass, VC: Vegetation cover.

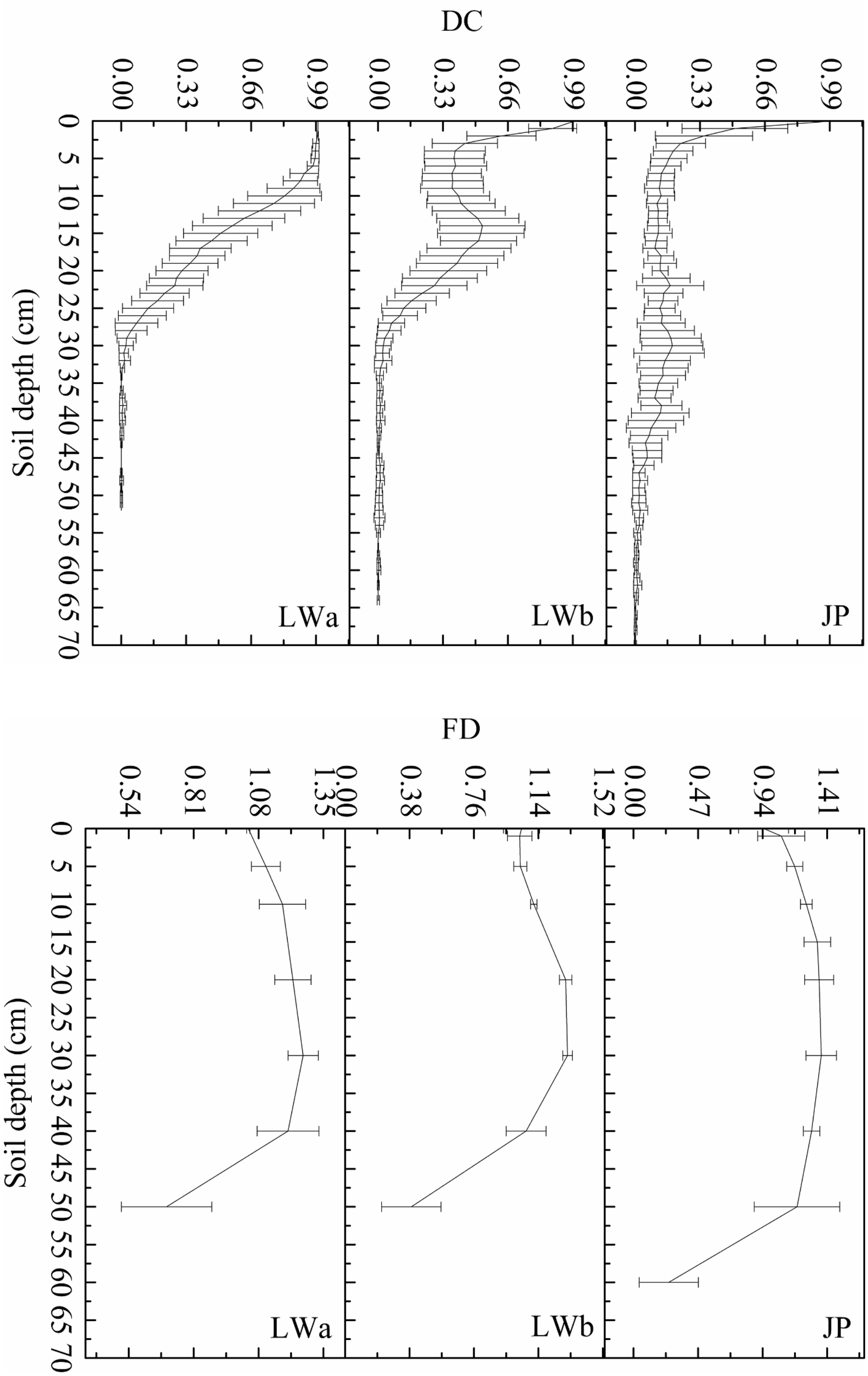

Fig. 2. Changes of *DC* and *FD* with increasing soil depth in the study areas.

2.4. IHC and wetland degradation assessment

2.4.1. IHC characterization

To show how the index of hydrological connectivity (*IHC*) can be used as a wetland degradation indicator, the available and historical data for the dye-stained areas in the study area were compiled for published articles, and the longitudes and latitudes of the sampling sites were marked in ArcGIS. Remote sensing image acquisition times between July and August in 2005, 2010, 2014, and 2020 were selected in the study area because *Phragmites australis* and *Suaeda glauca* could be easily distinguished in these months. Furthermore, the Landsat series remote sensing images of the study area from July to August in each selected year were screened. A series of pre-processing procedures, including radiation correction, cloud pollution removal, and image stitching, were performed (Cui et al., 2021). The *IHC* in the soil profiles during different periods was calculated based on the *DC* and *FD* collected from published papers. According to the changes in the dye-stained areas with increasing soil depth, as well as the corresponding change in the fractal dimensions in similar dye-stained areas (Fig. 3), a novel *IHC* was defined (Eq. (3)) to quantitatively characterize the changes in the hydrological connectivity both spatially and temporally.

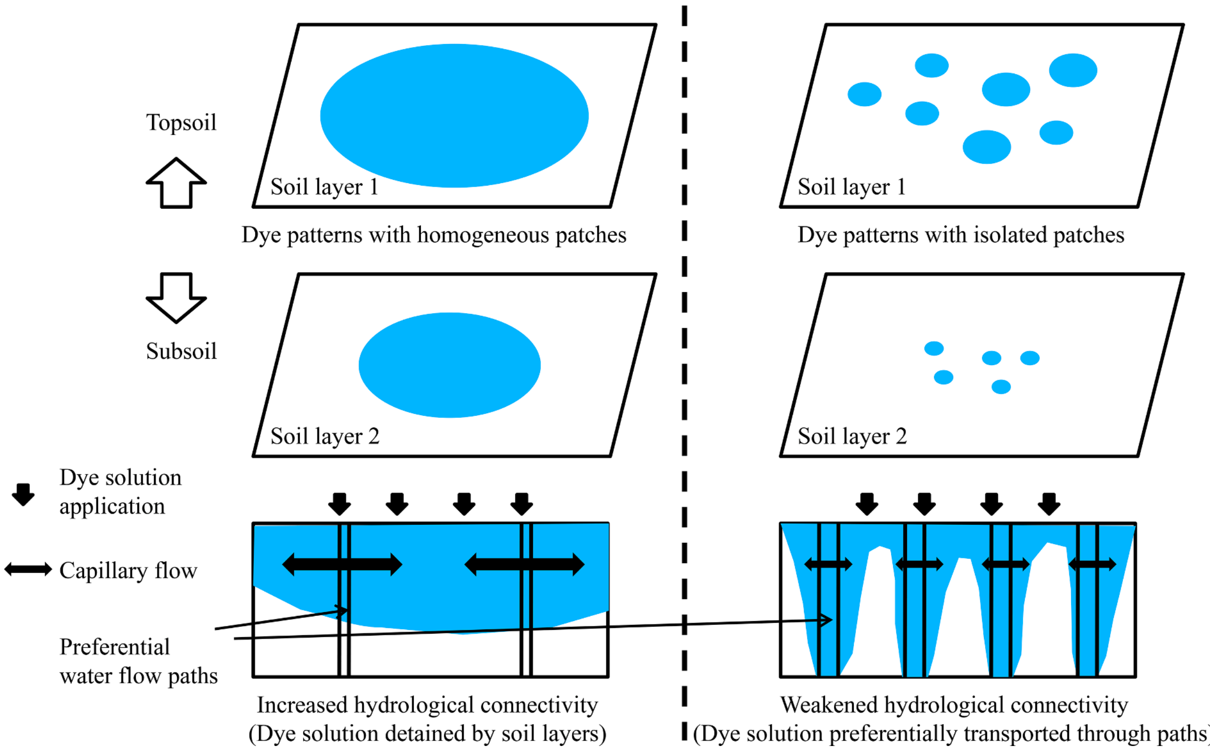

Fig. 3. Different dye patterns with different state of hydrological connectivity. The more homogeneous the dye patterns, the higher the hydrological connectivity at soil profiles. Large number of dye-stained areas and isolated stained patches showed low connectivity between the individual patch and the entire dye-stained areas.

$$IHC = \frac{\sum_{i=1}^{n} \frac{DC_i}{FD_i}}{n} \tag{3}$$

where *IHC* is the index of hydrological connectivity in the soil profiles, and *n* is the number of dye-stained horizontal profiles at a certain soil depth.

The value range is $0 < IHC < 1$. The larger the value of the *IHC* is, the better the wetland ecosystem services in the region are. If the *IHC* value was close to 1, the region was less affected and had more positive ecosystem services. It requires little time to balance the hydraulic pressure between the dye-stained areas and the non-dye-stained areas. If the *IHC* value was close to 0, the region had a weakened hydrological connectivity in the wetland ecosystems and more time was required to balance the hydraulic pressure between the dye-stained areas and the non-dye-stained areas. The dye solution was mainly transported through preferential water flow paths (Fig. 3) (Zhang et al., 2018).

2.4.2. Assessing wetland degradation

The 21 indexes of root parameters and soil properties were processed for the principal component analysis (PCA). The PCA was performed to group the root parameters and soil properties into statistical factors and to further establish the minimum data set (Zhao et al., 2019). The PCA was characterized by factor extraction and variable rotation. When more than one variable occurred under a single principal component, PCA was conducted to decrease redundancy (Mandal et al., 2017). Then, each indicator from the data set was transformed into a unit-less score ranging from 0 to 1. Since the principal components (PCs) with high eigenvalues were considered to represent the variations in the root parameters and soil properties, only the principal components with eigenvalues of > 1 were retained (Zhao et al., 2019). The communality of each root parameter and soil property was used to determine the proportion of the variance. For the root parameters (RL, RW, RSA, RPA, RV, and RB) and soil properties (SOM, STN, STP_1, SAP, STP_2, SNCP, SMHC, SCHC, SOWC, SWS, SV, SDC, SWC, SBD, and SEC), the communalities indicate which roots parameter or soil property explains $> 95\%$ of the variance. the components were identified using PCA. Each component explains a different variance. Generally, a component with a loading of > 0.6 is considered to be a major contributor (Zhao et al., 2019). A multiple regression model was established based on the principal components of the PCA to calculate the comprehensive score and to quantitatively describe the degree of wetland degradation in the study area.

2.5. Back propagation (BP) neural networks

In this study, back propagation (BP) neural networks computed using the Neuralnet package in R were used to predict the relationships between the different factors affecting the *IHC*. The BP neural network proposed by McClelland and Rumelhart (1985) is a multi-layer feedforward neural network, including an input layer, hidden layer, and output layer, which is trained according to the error reverse propagation algorithm. The structure of the BP neural network is shown in Fig. 4. The number of nodes in the input layer is 21. In the model, the number of nodes in the first hidden layer is 6, and that in the second hidden layer is 2. The number of nodes in the output layer is 1. To ensure the quality of a BP neural network, the input and output data should be normalized. The formula for normalization is as follows (Li et al., 2020; Sun and Huang, 2020):

$$Scalex_i = \frac{x_i - x_{min}}{x_{max} - x_{min}} \tag{4}$$

where X_{min} and X_{max} are the minimum and maximum values of the input matrix and output vector quantity, respectively, and X_i is the real value of each vector quantity.

The input value of the BP neural network is expressed as $X = (x_1, x_2, x_3..., x_n)$. The implicit layer excitation function f selected in this study is

$$f(U_j) = \frac{1}{1 + e^{-U_j}} \tag{5}$$

The output H_j of the hidden layer can be obtained as follows:

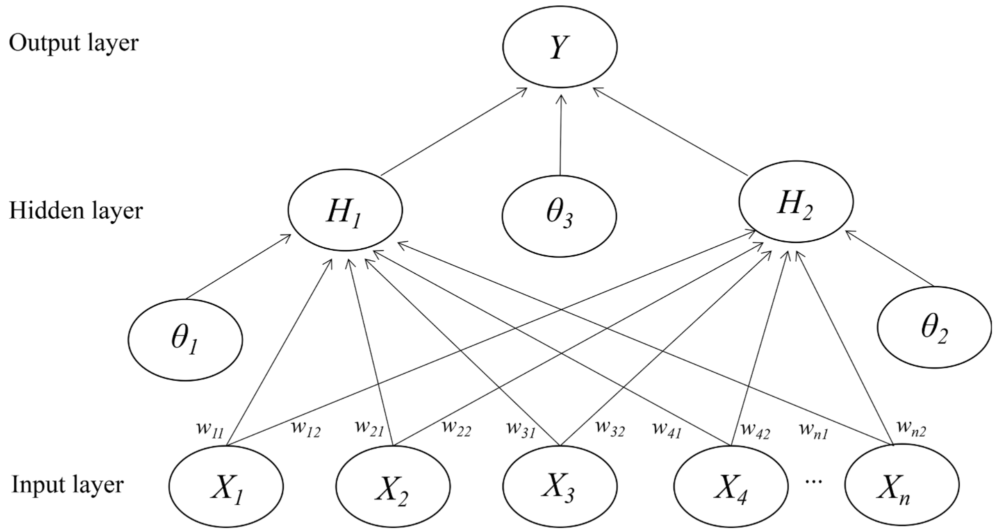

Fig. 4. The structure of the BP neural network.

$$H_j = f(U_j)j = 1, 2\cdots, l, \tag{6}$$

$$U_j = \sum_{i=1}^{n} w_{ij}x_i + \theta_j \tag{7}$$

where w_{ij} is the connection weight between the input later and the output layer, θ_j is the hidden layer threshold, f is the activation function of the hidden layer, U_j is the input of the hidden layer node, and l is the number of nodes in the hidden layer.

The predicted output O_k of the BP neutral network can be obtained as follows:

$$O_k = \sum_{i=1}^{l} H_j w_{jk} + \theta_k, \tag{8}$$

where w_{jk} is the connection weight. θ_k is the output layer threshold.

The goodness of fit (R^2) and the root mean square error (*RMSE*) of the BP model, which reflects the difference between the predicted output O_k and the expected output Y_k , are calculated as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(Y_k - O_k)^2}{\sum_{i=1}^{n}\left(Y_k - \overline{Y_k}\right)^2}, \tag{9}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(Y_k - O_k)^2} \tag{10}$$

In this manner, we aim to determine the relative importance of the different factors used to predict the *IHC*. The generalized weight (GW) of the BP neural network was adopted. The GW was calculated using the Neuralnet package in R in order to estimate the relative importance of the variables. The GW is calculated as follows:

$$GW_i = \frac{\partial(log\frac{O_k}{1-O_k})}{\partial x_i}. \tag{11}$$

2.6. Global sensitivity and uncertainty analyses

Global sensitivity and uncertainty analysis (GSUA) investigates how the variation in the output of a model can be attributed to the variations of in its different input factors (Pianosi et al., 2016; Xu et al., 2019). GSUA is a method that ranks the importance and interactions between the different input factors (Convertino et al., 2019). A fully probabilistic information theoretical GSUA that considers all of the probability distribution functions of the different factors was adopted. Following the most common procedure for evaluating the model, we constructed the receiver operating characteristic (ROC) curves for the BP neutral network model. The ROC curve plots the probability of detecting a sensitivity and 1-specificity for the entire range of possible cut-off points. We analyzed the areas under the ROC curves (AUCs). It is suggested that an AUC value of 0.50 indicates no ability to discriminate, while a value of $<$ 0.60 indicates poor discrimination. A value between 0.60 and 0.75 indicates possibly helpful discrimination, and a value of $>$ 0.75 reflects useful discrimination.

2.7. Statistical analysis

One-way analysis of variance (ANOVA) was used to compare the differences in the plant roots, soil properties, and *IHC* in each soil layer using the SPSS 22.0 software package. The Pearson correlation coefficient was calculated to test the effects of the plant root parameters and the soil properties on the *IHC*. The results are reported at a 5% significance level.

3. Results and discussion

3.1. IHC assessment in soil profiles in the study area

The *IHC* values of the different plots (Fig. 5) were compared. The comparisons between the investigated years revealed that the proportion of high *IHC* values in the study area increased during 2005–2020. In 2005, the high *IHC* values were concentrated in the middle of the study area. Since 2010, the *IHC* has been extremely fragmented, and it was dense, reaching gradients of 0.257 ± 0.127 and 0.392 ± 0.209. The weakened *IHC* areas were widely distributed in the eastern part of the study area. The western and middle wetland areas maintained high *IHC* values in the long term. With respect to the surface soils (0–10 cm), the *IHC* values were 0.609 ± 0.134, 0.399 ± 0.057, and 0.225 ± 0.163 in the LWa, LWb, and JP areas, respectively. For soil depths of 10–20 cm, the *IHC* values were 0.395 ± 0.100, 0.229 ± 0.071, and 0.098 ± 0.003 in the LWa, LWb, and JP areas, respectively. With respect to the deeper soils (20–30 cm), the *IHC* values were 0.172 ± 0.048, 0.142 ± 0.072, and 0.137 ± 0.027 in the LWa, LWb, and JP areas, respectively. For the rhizosphere (0–30 cm from the soil surface), the *IHC* values were 0.392 ± 0.209, 0.257 ± 0.127, and 0.153 ± 0.099 in the LWa, LWb, and JP areas, respectively. These results show that the hydrological connectivity in the LWa was always higher than that in the LWb and JP. This was because the presence of more litter and thick humus on the land surface in the LWa reduced the splash erosion and soil crusting. A higher

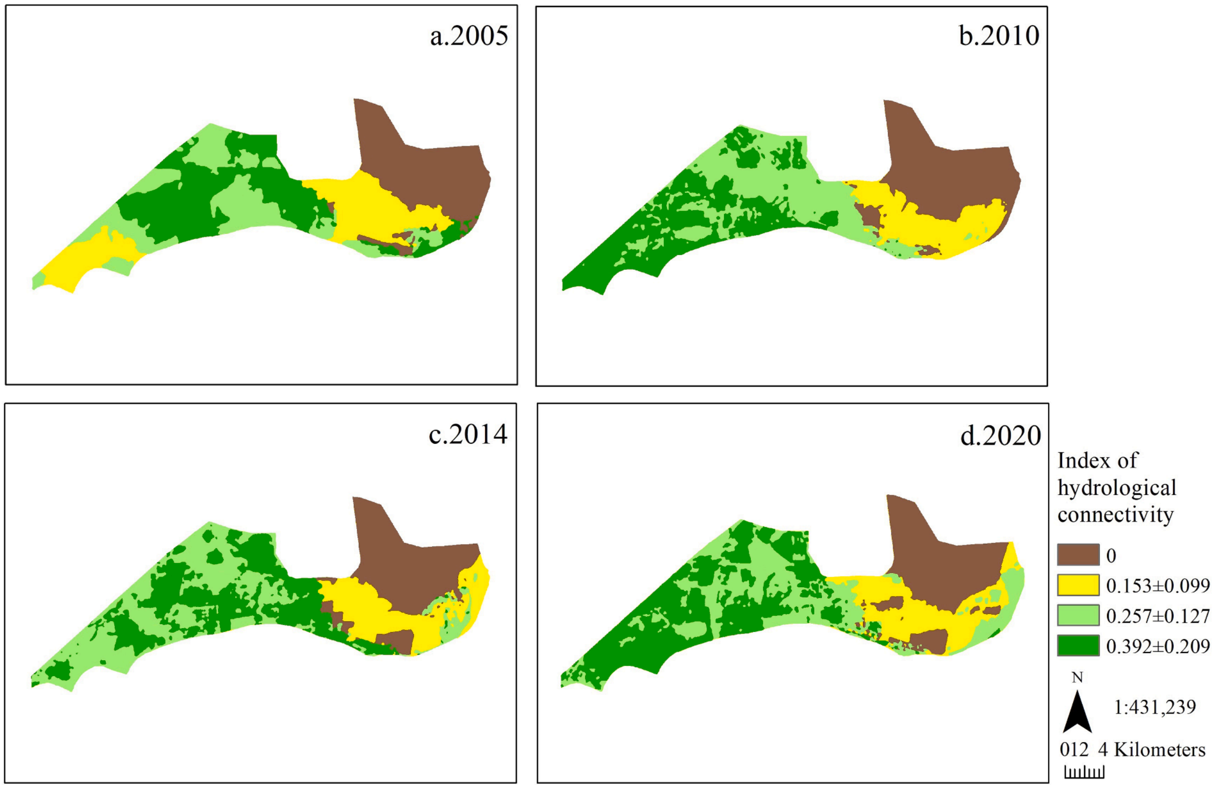

Fig. 5. The changes of the levels of *IHC* within the years.

abundance of roots, pore formation at the interface between the roots and the soil particles, and preferential channels formed by root decomposition improved the hydrological connectivity (Zhang et al., 2017, 2021). However, the LWb had less litter and humus on the land surface, which did not prevent the soil structure from being destroyed. The seriously compacted land surface in the LWb led to greater surface runoff instead of higher infiltration into the soil. In particular, the soil water content in the LWb was lower than that in the LWa (Table 1). The lower soil water content in the LWb may have caused obvious soil water repellency (Rodríguez-Alleres et al., 2012; Benito Rueda et al., 2016). Soil water repellency contributed more to the high spatial heterogeneity of the water flow paths, exhibiting complicated dyed patches and variable *IHC* values at each soil depth in the LWb (Alanís et al., 2017). In the JP, the presence of more geographically isolated patches of hydrological connectivity led to extremely complex hydrological responses compared with the LWa and LWb. The serious wetland degradation in the JP promoted the development of more soil fissures in soil profiles (Kim and Mohanty, 2017). As dense preferential flow paths, these soil fissures promoted the rapid transport of solutions from surface soils to the deep soils, which resulted in a weakened hydrological connectivity (Kim and Mohanty, 2017; Zhang et al., 2021).

3.2. Ability of the IHC to reflect wetland degradation

In the soil depth interval of 0–10 cm, the first and second principal components explained 73.84% and 26.16% of the original information, respectively, and the cumulative interpretation amount was 100%. The first principal component (PC1) had the highest correlation with the soil total porosity, soil capillary holding capacity, and soil ventilation and the lowest correlation with the soil bulk density and soil water storage. The second principal component (PC2) had the highest correlation with the root surface area and root projected area and the lowest correlation with the soil electrical conductivity. PC1 represents the soil information, and PC2 represents the root information. In the soil depth interval of 10–20 cm, PC1 and PC2 explained 88.64% and 11.36% of the original information, respectively. PC1 had the highest correlation with the root length and root volume and the lowest correlation with the soil water content and soil water storage. PC2 had the highest correlation with the soil organic matter and soil total nitrogen and the lowest correlation with the soil electrical conductivity. PC1 represents the root information, and PC2 represents the soil information. In the soil depth interval of 20–30 cm, PC1 and PC2 explained 67.40% and 32.60% of the original information, respectively. PC1 had the highest correlation with the root biomass, root surface area, and root projected area and the lowest correlation with the soil water content and soil water storage. PC2 had the highest correlation with the soil maximum holding capacity and optimum soil water content and the lowest correlation with the soil bulk density and soil electrical conductivity. PC1 represents the root information, and PC2 represents the soil information. The sorting scatter diagram of the 21 indexes of the root parameters and soil properties with PC1 and PC2 as the coordinate axes was plotted (Fig. 6a–c). Multiple regression models for assessing the degree of wetland degradation were established.

$$\text{PC (0–10 cm)} = 0.7384 \cdot \text{PC1} + 0.2616 \cdot \text{PC2}.$$
$$\text{PC (10–20 cm)} = 0.8864 \cdot \text{PC1} + 0.1136 \cdot \text{PC2}.$$
$$\text{PC (20–30 cm)} = 0.6740 \cdot \text{PC1} + 0.3260 \cdot \text{PC2}.$$

The results of PCs are shown in Fig. 7. Based on the field observations and PCA, the PC values were divided into three intervals: (1) $2.691 < \text{PC} < 3.827$ indicates a moderately degraded wetland (LWa); (2) $-0.009 < \text{PC} < 0.033$ indicates a severely degraded wetland (LWb); and (3) $-3.821 < \text{PC} < -2.682$ indicates an extremely degraded wetland (JP). Here, we introduced the *IHC* as an indicator of the status of wetland degradation. The characterization of the *IHC* indicates that the study area had comparable conditions, which confirms that the *IHC* is indeed a suitable indicator of wetland degradation. Theoretically, this novel assessment indicator of wetland degradation indicates that the wetland

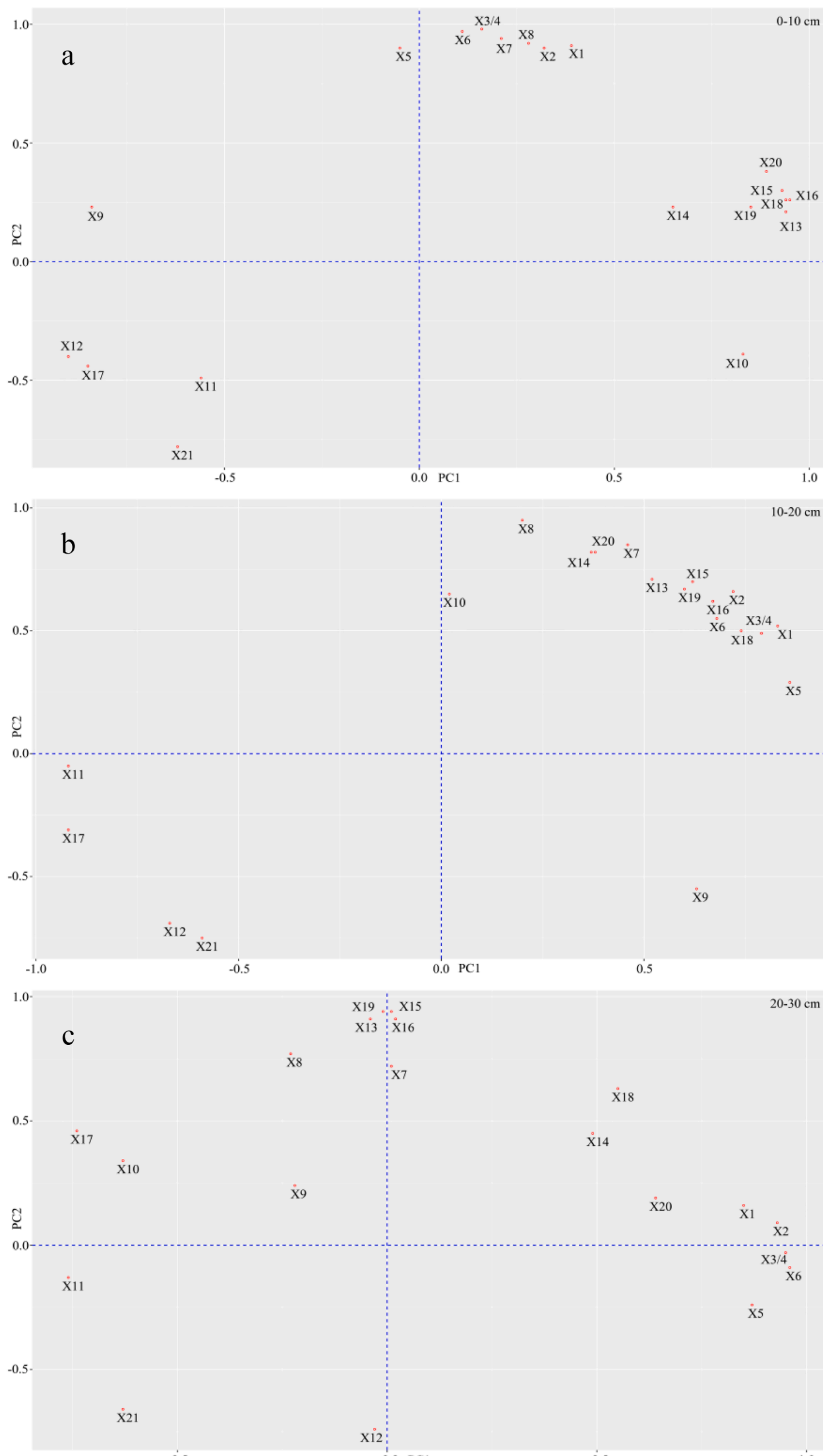

Fig. 6. The sorting scatter diagram of the 21 index of "roots parameters and soil properties" (X1: Root length, X2: Root width, X3: Root surface area, X4: Root projected area, X5: Root volume, X6: Root biomass, X7: Soil organic matter, X8: Soil total nitrogen, X9: Soil total phosphorus, X10: Soil available phosphorus, X11: Soil water content, X12: Soil bulk density, X13: Soil total porosity, X14: Soil non-capillary porosity, X15: Soil maximum holding capacity, X16: Soil capillary holding capacity, X17: Soil water storage, X18: Soil ventilation, X19: Soil optimum water content, X20: Soil drainage capacity, X21: Soil electronic conductivity).

degradation is more serious in the LWb and JP than in the LWa. As is shown in Fig. 7, the results indicate that the PC value of the wetland degradation is positively correlated with *IHC*. For each soil depth interval (0–10, 10–20, and 20–30 cm), in order of the PC value and *IHC*, the order of the three wetlands is LWa > LWb > JP. The high hydrological connectivity in soil profiles in the LWa indicates that the hydrological responses could be promoted (Zhang et al., 2021). The high hydrological responses in the LWa carried more energy and organisms, providing for more wetland ecosystem functions compared with the LWb and JP. This was because the high hydrological connectivity

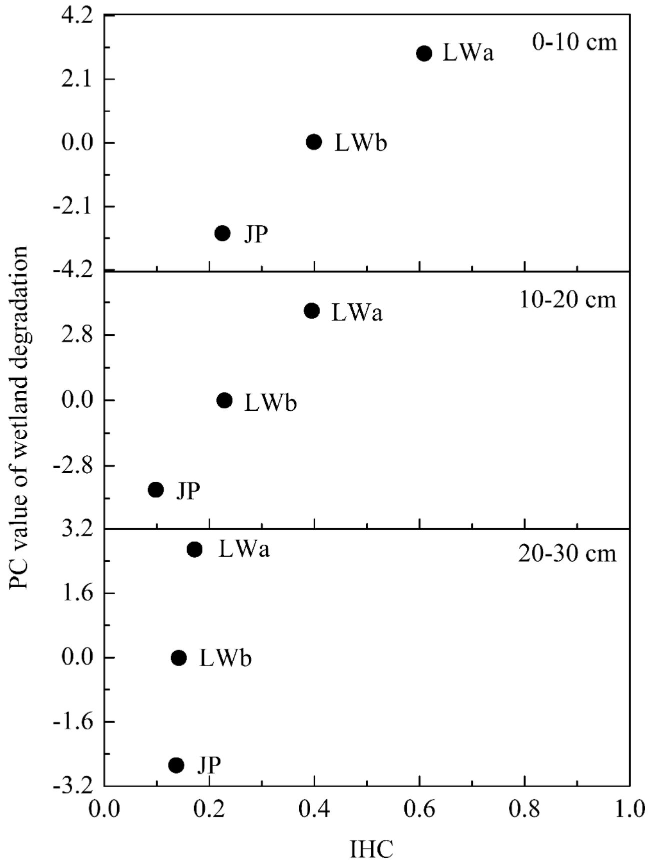

Fig. 7. Different *IHC* reflects different PC value of wetland degradation.

increased the efficiency of the mass transfer from one place to another (Zhang et al., 2021). In soil profiles, the high hydrological connectivity increased the soil water holding capacity and formed more connected horizontal or vertical paths (Zhang et al., 2021). However, the weakened hydrological connectivity in soil profiles located in the geographically isolated patches in the LWb and JP may have greatly shortened the retention time of the mass in the deeper soils (Zhu et al., 2020). The complex isolated patches in the LWb and JP enhanced the spatial heterogeneity of the hydrological connectivity (Zhu et al., 2020). The mass was transport through preferential water flow paths, which reduced the efficiency of the wetland treatment (Casey and Klaine, 2001; Parsons et al., 2006). The complicated preferential water flow paths in the LWb and JP resulted in weakened hydrological connectivity and ultimately in serious wetland degradation. Therefore, reestablishing hydrological connectivity through the construction of a seepage layer at soil depths of 0–10 cm in the LWb and JP would decrease the surface runoff generation (Wu et al., 2020). Shallow ploughing treatment in the LWb and JP would increase the topsoil's hydrological connectivity, which would decrease the connectivity between the surface water and groundwater and would also decrease the risk of groundwater contamination (Wu et al., 2020). Furthermore, adding litter to the surface soils of the LWb and JP would also improve the topsoil properties (Zhang et al., 2019). Appropriate freshwater input to the LWb and JP should be conducted to alleviate the water deficiency. Freshwater input would create a continuous horizontal subsurface flow in the wetland (Cui et al., 2009). The increase in the hydrological connectivity in soil profiles may contribute to the horizontal continuity, which prevents wetland clogging, thereby reducing the preferential water flow paths and increasing the sewage purification efficiency (Muñoz et al., 2006).

3.3. Relevance of the root-soil interface to the IHC

The *IHC* is based on the definition of neural networks that indicate how different factors are interdependently correlated with the *IHC*. The constructed BP neural networks (Fig. 8) display the strengths of each factor. The R^2 value of the BP neural networks was 0.47, and the *RMSE* was 0.72. Based on the results, the *IHC* in soil profiles is correlated with the root-soil interface that best explains the representative factors. As was previously explained, these factors were selected as those with a weighted in-degree higher than their weighted out-degree. Within the networks, we defined the GW useful for assessing the relative importance of the integrated factors. The GW of these factors (Fig. 9) indicates that the soil drainage capacity, soil maximum holding capacity, soil non-capacity porosity, soil optimum water content, and soil ventilation were more sensitive and important to the *IHC*. The results of this study indicate that the soil properties had a larger influence on the *IHC* than the root systems, which is consistent with the results of Carminati et al. (2017). Although the soil properties played the most significant role in the hydrological connectivity compared with the root systems, the effects of the root systems cannot be neglected. This is because the hydrological responses of the root-soil interface, which is dominated by the root systems, could also play a crucial role in the water balance (Volpe et al., 2013). For example, the decaying or decayed roots increase the soil organic matter content and promote the formation of soil aggregates (Cui et al., 2019; Liu et al., 2020; Wu et al., 2021). The formation of soil aggregates can improve the soil hydrological responses (Cui et al., 2019; Wu et al., 2021). Some studies have also concluded that root systems are positively correlated with increased hydrological responses. This is because roots can improve the soil structure (Wu et al., 2016; Wu et al., 2021). Root systems can not only affect the hydrological responses by changing soil properties, but they can also improve the hydrological responses by forming preferential root channels (Wu et al., 2021). Complex root systems promote the formation of large pores and form root channels for mass transfer. Root channels increase the density of large pores and improve the connectivity of these pores, which may be more conductive to mass transfer (Bogner et al., 2010; Wu et al., 2021). However, in wet soils, some studies have concluded that the hydrological responses of root systems can be neglected compared with those of soils (Gardner, 1965), and the hydrological conductivity of wet soils is higher than that of the root systems (Carminati et al., 2017). The variations in the hydrological responses of root systems are based on the principles of mass transfer into a network of narrow pipes. Little is known about the hydrological responses of the entire root systems at the level of the individual root systems (Draye et al., 2010). In wet soils, more living root systems than decayed or decaying root systems are distributed in the soil layers, whereas in dry soils, more dead root systems than living root systems are distributed in the soil layers (Ludovici, 2004). Decomposition of root systems provides a continuous network of roots channels and ultimately improves the soil's porosity. However, in wet soils, the hydrological responses of wet soils could be impeded due to the mucilage from root exudates (Benard et al., 2016; Carminati et al., 2017). In dry soils, the hydrological responses of soils significantly decrease, becoming the limiting factor for the hydrological responses of roots (Draye et al., 2010). In dry soils, a local decrease in the soil water potential leads to a drop in the soil conductivity, which restrains the mass transfer. Root shrinkage in dry soils results in a loss of contact with the surrounding soils and ultimately reduces the hydrological responses of the root-soil interface (Faiz and Weatherley, 1982). Loss of physical contact with the soil particles promotes the formation of air-filled gaps at the root-soil interface when roots shrink (Carminati et al., 2017). As more soil spaces become filled with air, there are few continuous channels through which mass can be easily transferred (Ludovici, 2004). Under intermediate conditions, the hydrological responses of soils and roots can be comparable (Draye et al., 2010). In fact, the hydrological responses of roots are low due to the limiting soil factors (Draye et al., 2010). This is because root exudates become hydrophobic under

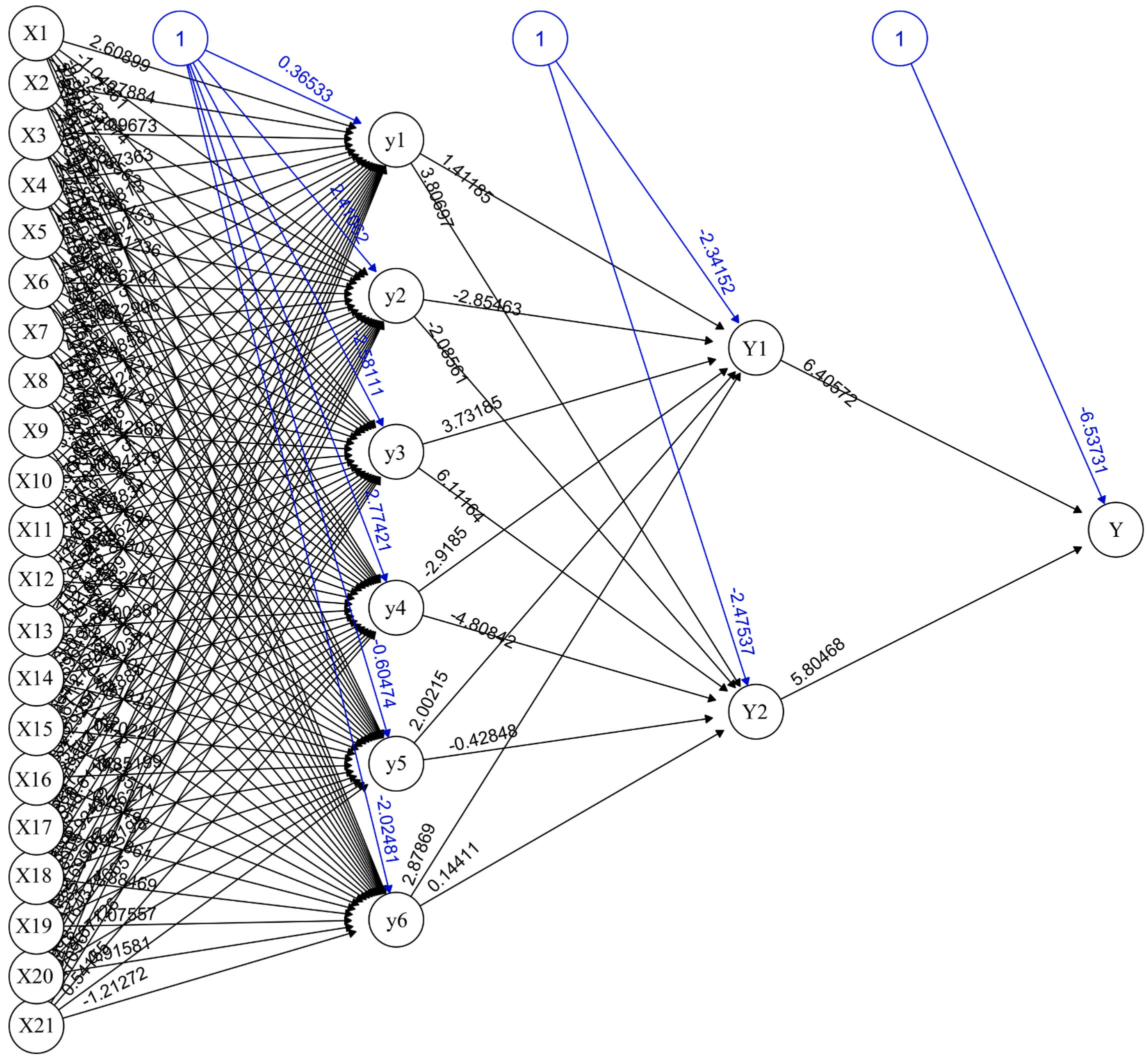

Fig. 8. Visual BP neural network from the input layer to the output layer.

drought conditions, which limits the rewetting of the soils (Benard et al., 2016). The rhizosphere is still dry and will be rewetted a few days after the bulk soil rewetting due to the effects of root mucilage (Carminati et al., 2017). Therefore, to better improve the interactions between the roots and soil, wetland managers should also adjust the frequency and depth of freshwater replenishment to increase the soil water content and the soil organic matter and to reduce the hydrophobicity of roots during reed growth in the dry season.

The results of the systemic sensitivity analysis are shown in Fig. 10. In particular the ROC is needed to measure the prediction accuracy. Sixty-eight percent of the observed *IHC* are reproduced, which translates into an AUC of 0.68. The ROC curve shows that the weakened hydrological connectivity is not reproduced well due to the small values of the *IHC*. The ROC curve shows an increase in sensitivity (Convertino et al., 2019). The AUC is more applicable to the generalizability of the current BP neural networks. While the AUC measures the BP model's performance, it is important for wetland managers to conduct a series of wetland restoration measures. Choosing an appropriate cut-off value is important for *IHC* prediction. Several criteria based on the ROC curve have been proposed for selecting the most appropriate cut-off value. The ROC curve shows that it is possible for BP neural networks to obtain a higher sensitivity. The BP neural networks have high reliability regarding predictions, which presents new possibilities for wetland managers in terms of degraded wetland restoration.

The results show that the soil holding capacity, soil non-capacity porosity, and soil ventilation were comparably important to the hydrological connectivity in soil profiles. The ability of the soil holding capacity could reflect the types of soil texture. Several studies have shown that clay soils contain more small soil pores which are homogeneously distributed in soil profiles. These homogeneous connected pores could promote hydrological responses (Zhang et al., 2017, 2021). The soil non-capacity porosity, which reflects the ability of water to flow into soils driven by gravity, could increase the soil infiltration. The non-capacity pores could also promote the development of vertically and horizontally connected paths (Zhang et al., 2017). Therefore, wetland managers should effectively increase the soil holding capacity through improving the soil texture. As organic materials, biochar and straw should be used in the study area to improve the soil structure. Furthermore, managers should fully consider the functions of the root systems since they could decrease the soil compaction and improve the soil holding capacity, which could ultimately improve the hydrological responses.

4. Conclusions

The results show that the *IHC* significantly changed both spatially and temporally. The high hydrological connectivity in soil profiles was concentrated in the middle and western parts of the study area. The spatial distribution of the high hydrological connectivity has been extremely fragmented since 2010. The LWb and JP were more seriously

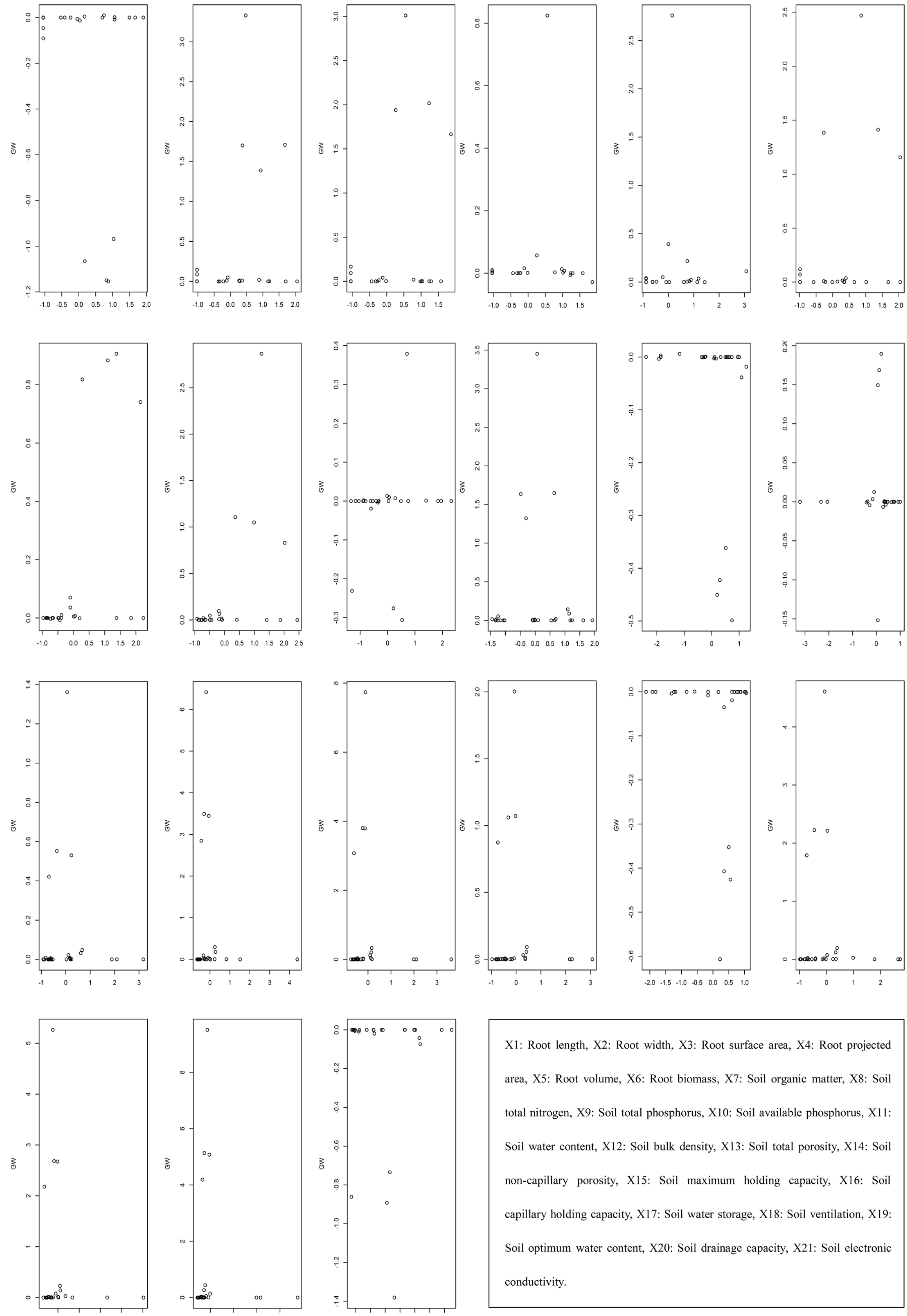

Fig. 9. The GW of different input factors contribution to the predicted output *IHC*.

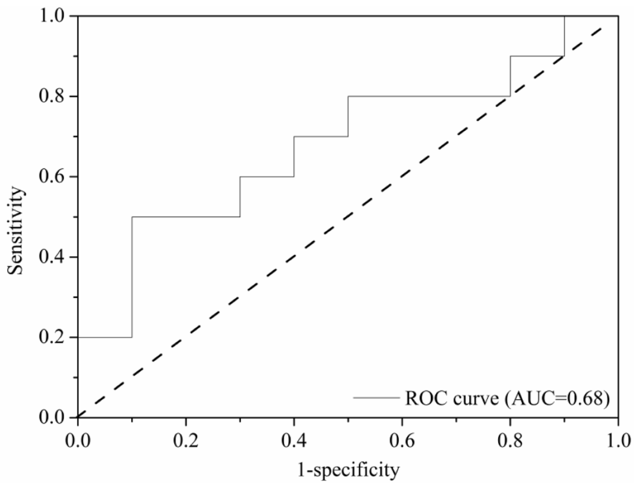

Fig. 10. ROC curve for the BP neural networks.

degraded than the LWa in the study area. The changes in the *IHC* were positively correlated with the PC value of the wetland degradation. The *IHC* is a novel indicator of the status of wetland degradation. Furthermore, the soil holding capacity, soil non-capacity porosity, and soil ventilation are relatively important to the changes in the *IHC*. Compared with the soil properties, the hydrological responses of the root systems can be neglected at the root-soil interface.

Understanding the hydrological connectivity in soil profiles is the first step of wetland restoration and conservation. Based on the results of this study, we propose an alternative wetland restoration solution for the Yellow River Delta: 1) a seepage layer in the surface soils (0–10 cm), shallow ploughing treatment, and litter or straw return to the soil surface should be conducted to increase the soil surface hydrological connectivity; 2) reeds should not be reaped every year to remove the nutrients from this area; and 3) the appropriate freshwater input should be strengthened.

CRediT authorship contribution statement

Yinghu Zhang: Data curation, Formal analysis, Investigation, Methodology, Writing – original draft. **Jinhong Chen:** Investigation, Software. **Jinchi Zhang:** Methodology. **Zhenming Zhang:** Investigation, Methodology. **Mingxiang Zhang:** Data curation, Investigation, Methodology, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Alanís, N., Hernández-Madrigal, V.M., Cerdà, A., Muñoz-Rojas, M., Zavala, L.M., Jordán, A., 2017. Spatial gradients of intensity and persistence of soil water repellency under different forest types in central Mexico. Land Degrad. Dev. 28 (1), 317–327.

Bai, J., Yu, Z., Yu, L.u., Wang, D., Guan, Y., Liu, X., Gu, C., Cui, B., 2019. In-situ organic phosphorus mineralization in sediments in coastal wetlands with different flooding periods in the yellow river delta, china. Sci. Total Environ. 682, 417–425.

Benard, P., Kroener, E., Vontobel, P., Kaestner, A., Carminati, A., 2016. Water percolation through the root-soil interface. Adv. Water Resour. 95, 190–198.

Benito Rueda, E., Rodríguez-Alleres, M., Varela Teijeiro, E., 2016. Environmental factors governing soil water repellency dynamics in a Pinus pinaster plantation in NW Spain. Land Degrad. Dev. 27 (3), 719–728.

Bogner, C., Gaul, D., Kolb, A., et al., 2010. Investigating flow mechanisms in a forest soil by mixed-effects modelling. Eur. J. Soil Sci. 61 (6), 1079–1090.

Bremner, J.M., 1960. Determination of nitrogen in soil by the kjeldahl method. J. Agric. Sci. 55 (1), 11–33.

Carminati, A., Benard, P., Ahmed, M.A., Zarebanadkouki, M., 2017. Liquid bridges at the root-soil interface. Plant Soil 417 (1-2), 1–15.

Casey, R.E., Klaine, S.J., 2001. Nutrient attenuation by a riparian wetland during natural and artificial runoff events. J. Environ. Qual. 30 (5), 1720–1731.

Castello, L., Macedo, M.N., 2016. Large-scale degradation of Amazonian freshwater ecosystems. Glob. Change Biol. 22 (3), 990–1007.

Chen, A., Sui, X., Wang, D., Liao, W., Ge, H., Tao, J., 2016. Landscape and avifauna changes as an indicator of Yellow River Delta Wetland restoration. Ecol. Eng. 86, 162–173.

Convertino, M., Annis, A., Nardi, F., 2019. Information-theoretic portfolio decision model for optimal flood management. Environ. Modell. Software 119, 258–274.

Cui, G., Liu, Y., Tong, S., 2021. Analysis of the causes of wetland landscape patterns and hydrological connectivity changes in Momoge National Nature Reserve based on the Google Earth Engine Platform. Arab. J. Geosci. 14 (3), 1–16.

Cui, Z., Wu, G.-L., Huang, Z.E., Liu, Y.U., 2019. Fine roots determine soil infiltration potential than soil water content in semi-arid grassland soils. J. Hydrol. 578, 124023. https://doi.org/10.1016/j.jhydrol.2019.124023.

Cui, B., Yang, Q., Yang, Z., Zhang, K., 2009. Evaluating the ecological performance of wetland restoration in the Yellow River Delta, China. Ecol. Eng. 35 (7), 1090–1103.

Deng, X., 2020. Influence of water body area on water quality in the southern Jiangsu Plain, eastern China. J. Cleaner Prod. 254, 120136. https://doi.org/10.1016/j.jclepro.2020.120136.

Draye, X., Kim, Y., Lobet, G., et al. (2010). Model-assisted integration of physiological and environmental constraints affecting the dynamic and spatial patterns of root water uptake from soils. Journal of Experimental Botany, 61(8), 2145-2155.

Faiz, S.M.A., Weatherley, P.E., 1982. Root contraction in transpiring plants. New Phytol. 92 (3), 333–343.

Feng, J., Liang, J., Li, Q., Zhang, X., Yue, Y.i., Gao, J., 2021. Effect of hydrological connectivity on soil carbon storage in the yellow river delta wetlands of China. Chinese Geograph. Sci. 31 (2), 197–208.

Gardner, W.R., 1965. Dynamic aspects of soil water availability to plants. Annu. Rev. Plant Physiol. 16 (1), 323–342.

Guo, L., Liu, Y., Wu, G.L., et al., 2019. Preferential water flow: influence of alfalfa (medicago sativa l.) decayed root channels on soil water infiltration. J. Hydrol. 578, 124019.

Jiang, X.J., Chen, C., Zhu, X., Zakari, S., Singh, A.K., Zhang, W., Zeng, H., Yuan, Z.-Q., He, C., Yu, S., Liu, W., 2019. Use of dye infiltration experiments and HYDRUS-3D to interpret preferential flow in soil in a rubber-based agroforestry systems in Xishuangbanna, China. Catena 178, 120–131.

Kim, J., Mohanty, B.P., 2017. A physically based hydrological connectivity algorithm for describing spatial patterns of soil moisture in the unsaturated zone. J. Geophys. Res. Atmos. 122 (4), 2096–2114.

Li, X., Cheng, X.i., Wu, W., Wang, Q., Tong, Z., Zhang, X., Deng, D., Li, Y., 2020. Forecasting of bioaerosol concentration by a Back Propagation neural network model. Sci. Total Environ. 698, 134315. https://doi.org/10.1016/j.scitotenv.2019.134315.

Li, B., Hu, Y., Chang, Y.u., Liu, M., Wang, W., Bu, R., Shi, S., Qi, L.i., 2021b. Analysis of the factors affecting the long-term distribution changes of wetlands in the Jing-Jin-Ji region, China. Ecol. Indic. 124, 107413. https://doi.org/10.1016/j.ecolind.2021.107413.

Li, Z., Sun, W., Chen, H., Xue, B., Yu, J., Tian, Z., 2021a. Interannual and seasonal variations of hydrological connectivity in a large shallow wetland of north china estimated from landsat 8 images. Remote Sens. 13 (6), 1214. https://doi.org/10.3390/rs13061214.

Liu, J., Engel, B.A., Dai, L., Wang, Y.u., Wu, Y., Yan, G., Cong, L., Zhai, J., Zhang, Z., Zhang, M., 2019. Capturing hydrological connectivity structure of wetlands with indices based on graph theory: a case study in Yellow River Delta. J. Cleaner Prod. 239, 118059. https://doi.org/10.1016/j.jclepro.2019.118059.

Liu, Y.u., Guo, L., Huang, Z.e., López-Vicente, M., Wu, G.-L., 2020. Root morphological characteristics and soil water infiltration capacity in semi-arid artificial grassland soils. Agric. Water Manage. 235, 106153. https://doi.org/10.1016/j.agwat.2020.106153.

Lu, G., Xie, B., Cagle, G.A., Wang, X., Han, G., Wang, X., Hou, A., Guan, B.o., 2021. Effects of simulated nitrogen deposition on soil microbial community diversity in coastal wetland of the Yellow River Delta. Sci. Total Environ. 757, 143825. https://doi.org/10.1016/j.scitotenv.2020.143825.

Ludovici, K.H., 2004. Soil biology and tree growth | tree roots and their interaction with soil. Encyclopedia For. Sci. 1195–1201.

Mandal, U.K., Sharma, K.L., Venkanna, K., et al. (2017). Sustaining soil quality, resilience and critical carbon level under different cropping systems in semi-arid tropical alfisol soils. Current Science, 112(9), 1882.

McClelland, J.L., Rumelhart, D.E., 1985. Distributed memory and the representation of general and specific information. J. Exp. Psychol. Gen. 114 (2), 159–188.
Meng, B.o., Liu, J.-L., Bao, K., Sun, B., 2019. Water fluxes of nenjiang river basin with ecological network analysis: conflict and coordination between agricultural development and wetland restoration. J. Cleaner Prod. 213, 933–943.
Muñoz, P., Drizo, A., Cully Hession, W., 2006. Flow patterns of dairy wastewater constructed wetlands in a cold climate. Water Res. 40 (17), 3209–3218.
Olsen, S.R., Cole, C.V., Watanabe, F.S., et al., 1954. Estimation of available phosphorus in soils by extraction with sodium bicarbonate. Government Print Office, Washington, DC, pp. 1–19.
Parsons, A.J., Brazier, R.E., Wainwright, J., Powell, D.M., 2006. Scale relationships in hillslope runoff and erosion. Earth Surf. Proc. Land. 31 (11), 1384–1393.
Pianosi, F., Beven, K., Freer, J., Hall, J.W., Rougier, J., Stephenson, D.B., Wagener, T., 2016. Sensitivity analysis of environmental models: a systematic review with practical workflow. Environ. Modell. Software 79, 214–232.
Rodríguez, F., Mayor, Á.G., Rietkerk, M., Bautista, S., 2018. A null model for assessing the cover-independent role of bare soil connectivity as indicator of dryland functioning and dynamics. Ecol. Ind. 94, 512–519.
Rodríguez-Alleres, M., Varela, M.E., Benito, E., 2012. Natural severity of water repellency in pine forest soils from NW Spain and influence of wildfire severity on its persistence. Geoderma 191, 125–131.
Saco, P.M., Rodríguez, J.F., Moreno-de las Heras, M., Keesstra, S., Azadi, S., Sandi, S., Baartman, J., Rodrigo-Comino, J., Rossi, M.J., 2020. Using hydrological connectivity to detect transitions and degradation thresholds: applications to dryland systems. Catena 186, 104354. https://doi.org/10.1016/j.catena.2019.104354.
Servadio, J.L., Convertino, M., 2018. Optimal information networks: Application for data-driven integrated health in populations. Sci. Adv. 4 (2), e1701088.
Shen, M., Liu, X., 2021. Assessing the effects of lateral hydrological connectivity alteration on freshwater ecosystems: a meta-analysis. Ecol. Ind. 125, 107572. https://doi.org/10.1016/j.ecolind.2021.107572.
SIMS, James.R., Haby, Vincent.A., 1971. Simplified colorimetric determination of soil organic matter. Soil Sci. 112 (2), 137–141.
Sun, W., Huang, C., 2020. A carbon price prediction model based on secondary decomposition algorithm and optimized back propagation neural network. J. Cleaner Prod. 243, 118671. https://doi.org/10.1016/j.jclepro.2019.118671.
Turnbull, L., Wainwright, J., 2019. From structure to function: Understanding shrub encroachment in drylands using hydrological and sediment connectivity. Ecol. Ind. 98, 608–618.
Volpe, V., Marani, M., Albertson, J.D., Katul, G., 2013. Root controls on water redistribution and carbon uptake in the soil-plant system under current and future climate. Adv. Water Resour. 60, 110–120.
Wu, H., Gao, X., Wu, M., Zhu, Y.i., Xiong, R., Ye, S., 2020. The efficiency and risk to groundwater of constructed wetland system for domestic sewage treatment-A case study in Xiantao, China. J. Clean. Product. 277, 123384. https://doi.org/10.1016/j.jclepro.2020.123384.
Wu, G.-L., Yang, Z., Cui, Z., Liu, Y.u., Fang, N.-F., Shi, Z.-H., 2016. Mixed artificial grasslands with more roots improved mine soil infiltration capacity. J. Hydrol. 535, 54–60.
Wu, G.-L., Cui, Z., Huang, Z.e., 2021. Contribution of root decay process on soil infiltration capacity and soil water replenishment of planted forestland in semi-arid regions. Geoderma 404, 115289. https://doi.org/10.1016/j.geoderma.2021.115289.
Xu, X., Gao, P., Zhu, X., Guo, W., Ding, J., Li, C., Zhu, M., Wu, X., 2019. Design of an integrated climatic assessment indicator (ICAI) for wheat production: a case study in Jiangsu Province, China. Ecol. Ind. 101, 943–953.
Yang, Q., Hu, P., Wang, J., Zeng, Q., Yang, Z., Liu, H., Dong, Y., 2021. The stereoscopic spatial connectivity of wetland ecosystems: Evaluation method and regulation measures. Hydrol. Process. 35 (5) https://doi.org/10.1002/hyp.v35.510.1002/hyp.14074.
Yi, J., Qiu, W., Hu, W., Zhang, H., Liu, M., Zhang, D., Wu, T., Tian, P., Jiang, Y., 2020. Effects of cultivation history in paddy rice on vertical water flows and related soil properties. Soil Tillage Res. 200, 104613. https://doi.org/10.1016/j.still.2020.104613.
Zhang, G., Bai, J., Xi, M., Zhao, Q., Lu, Q., Jia, J., 2016. Soil quality assessment of coastal wetlands in the yellow river delta of china based on the minimum data set. Ecol. Ind. 66, 458–466.
Zhang, Y., Li, W., Sun, G.e., King, J.S., 2019. Coastal wetland resilience to climate variability: A hydrologic perspective. J. Hydrol. 568, 275–284.
Zhang, Y., Niu, J., Zhang, M., Xiao, Z., Zhu, W., 2017. Interaction between plant roots and soil water flow in response to preferential flow paths in northern China. Land Degrad. Dev. 28 (2), 648–663.
Zhang, Y., Zhang, Z., Ma, Z., Chen, J., Akbar, J., Zhang, S., Che, C., Zhang, M., Cerdà, A., Lupwayi, N., 2018. A review of preferential water flow in soil science. Can. J. Soil Sci. 98 (4), 604–618.
Zhang, Y., Huang, C., Zhang, W., Chen, J., Wang, L.u., 2021. The concept, approach, and future research of hydrological connectivity and its assessment at multiscales. Environ. Sci. Pollut. Res. 28 (38), 52724–52743.
Zhao, Q., Bai, J., Gao, Y., Zhao, H., Huang, Y., Zhang, W., Wang, J., Chen, G., 2019. Effects of freshwater inputs on soil quality in the Yellow River Delta, China. Ecol. Ind. 98, 619–626.
Zhu, J., Wang, X., Zhang, Q., Zhang, Y., Liu, D., Cai, A., Zhang, X., 2020. Assessing wetland sustainability by modeling water table dynamics under climate change. J. Cleaner Prod. 263, 121293. https://doi.org/10.1016/j.jclepro.2020.121293.